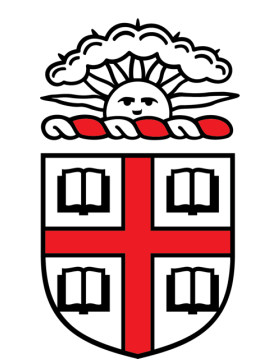


S²MoE: Spiking-based Mixture of Expert for Scientific Machine Learning

Dongyan Sun¹, Donghyun Lee¹, Bumjoon Kyle Rhee¹, Ruyin Wan², Sanjith T², Qian Zhang², George Em Karniadakis², Panos Stinis³, Priyadarshini Panda¹

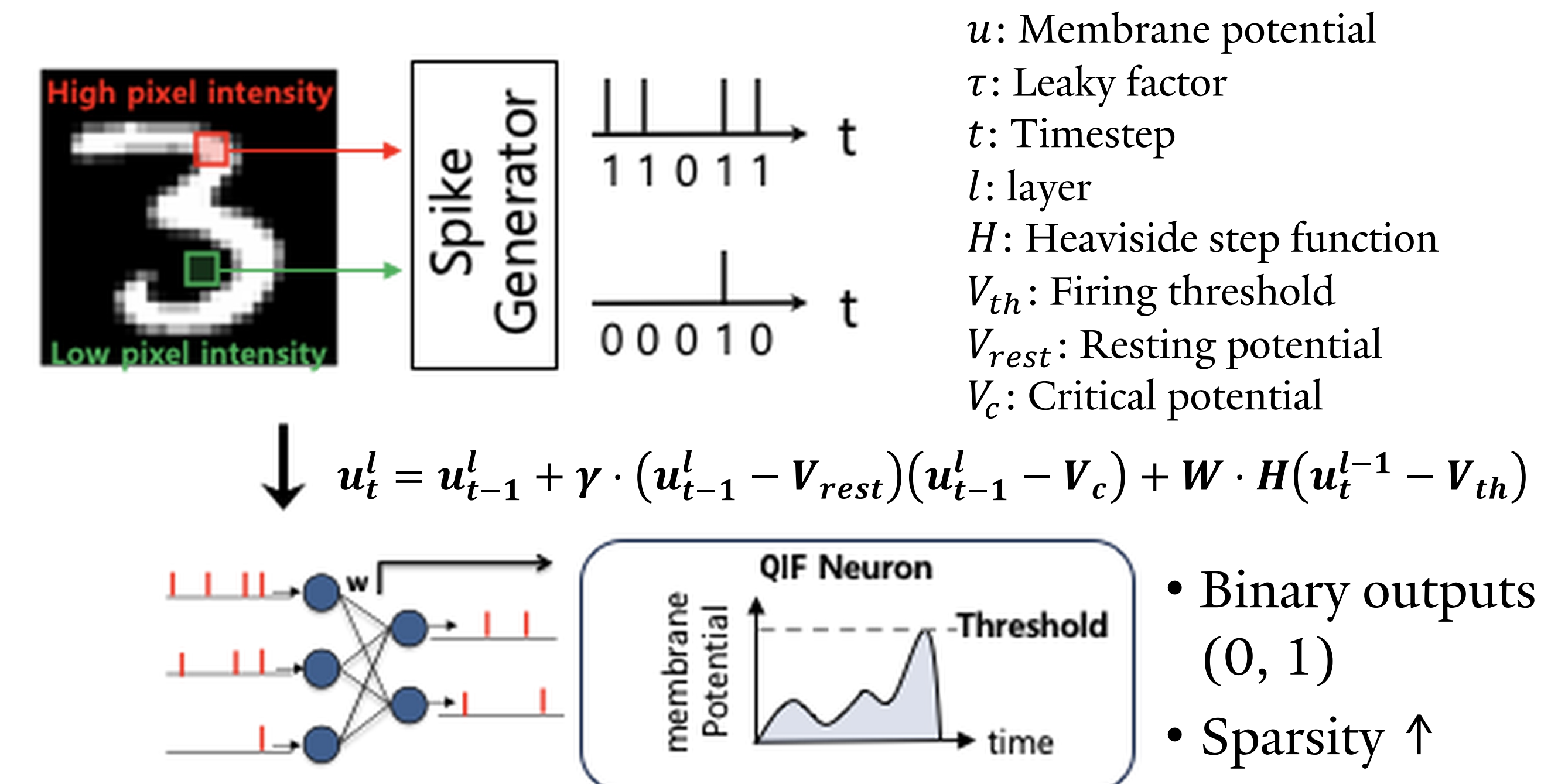
¹University of Southern California, ²Brown University, ³Pacific Northwest National Laboratory



BROWN



SPIKING NEURAL NETWORK



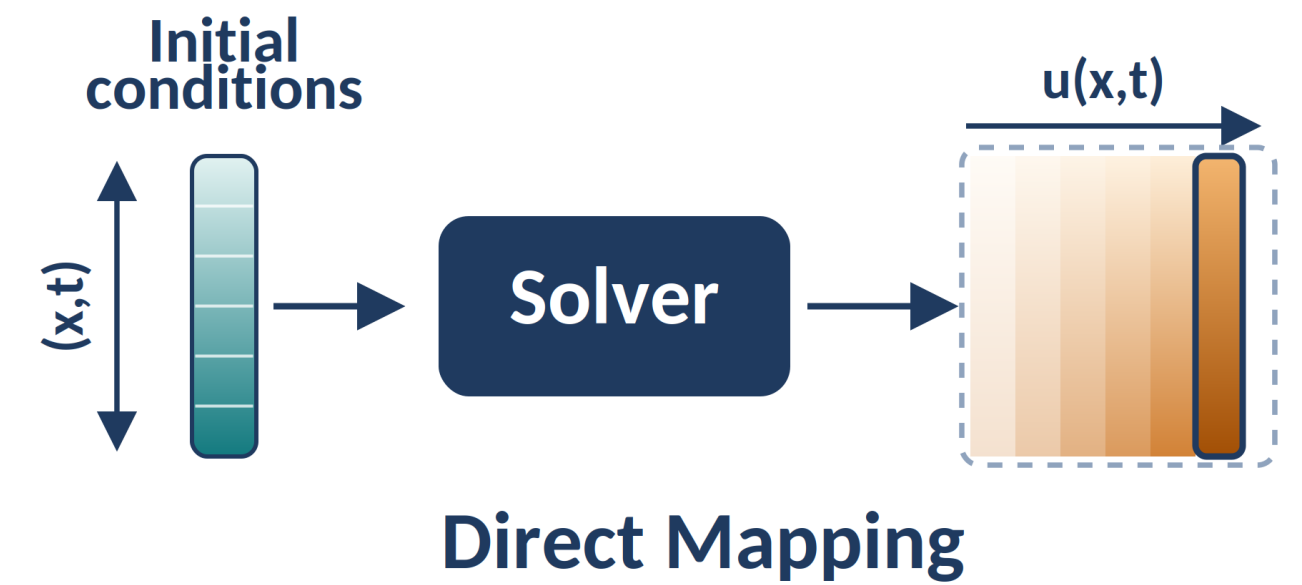
Quadratic Integrate-and-Fire (QIF)

- Quadratic membrane $\rightarrow$ higher representation power than linear (LIF) neurons
- Binary spike outputs preserve SNN sparsity & efficiency

ML-BASED PDE SOLVER

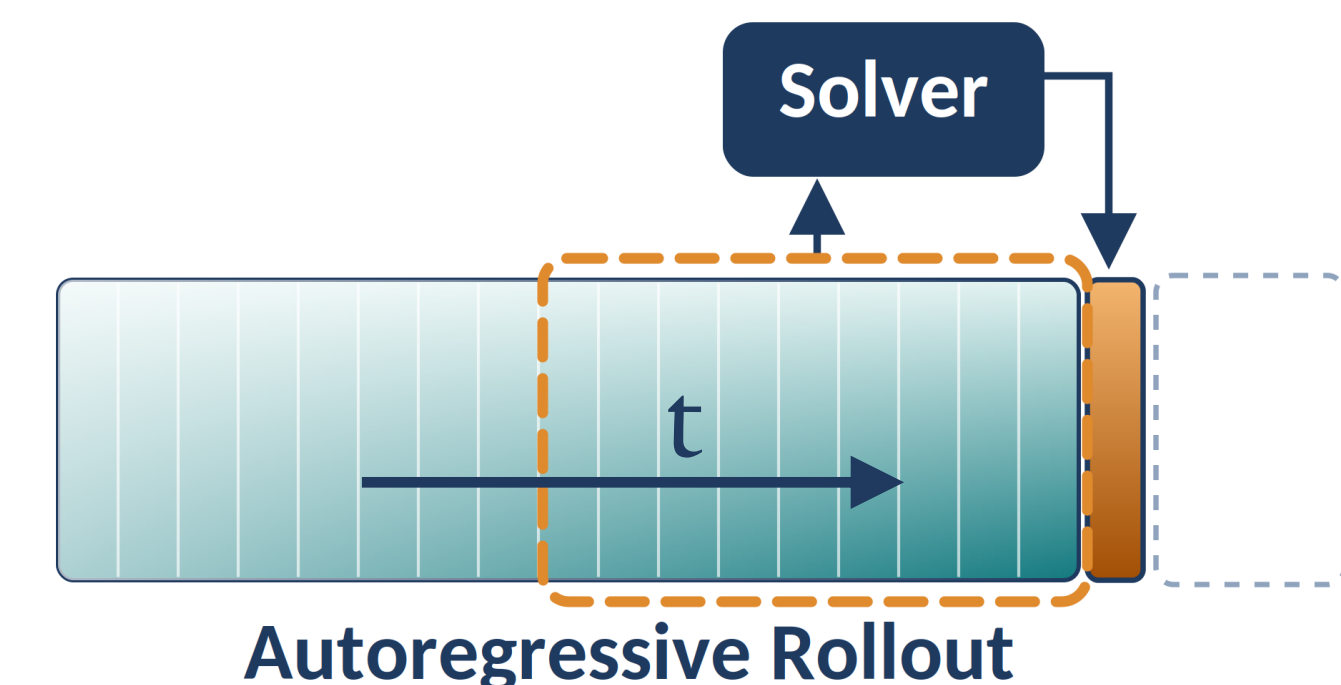
Direct Mapping

- Predict directly from coordinate to solution
- Redundant recomputation
- Wastes the membrane recurrence



Autoregressive Rollout

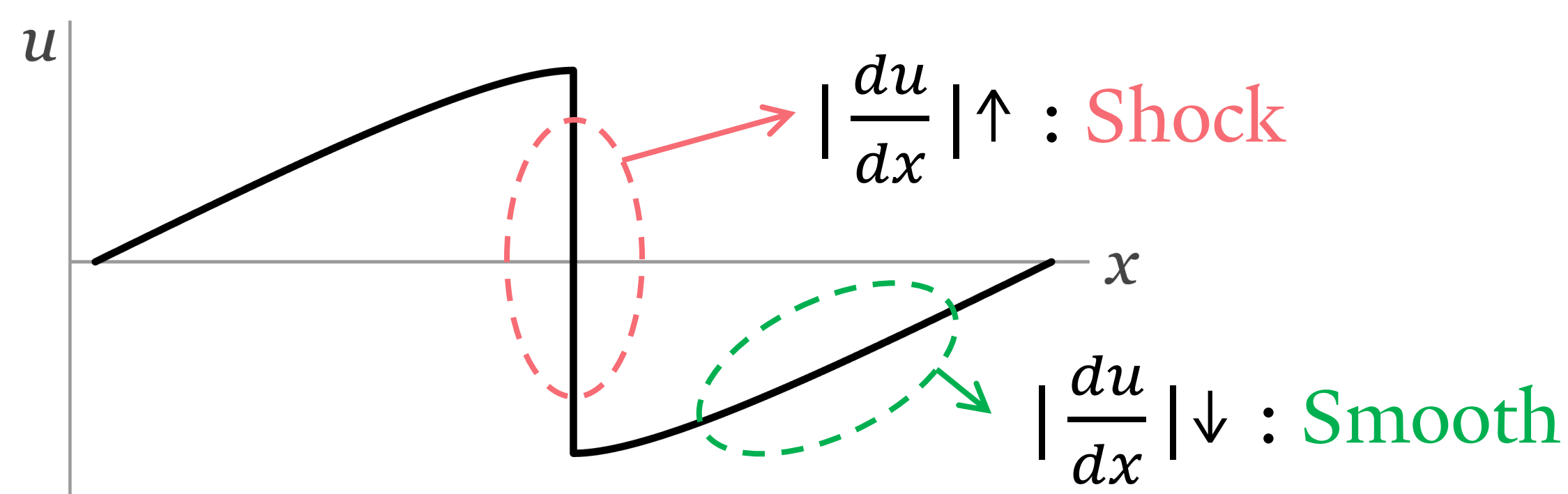
- Predict solution for next timestep reusing current solution
- Membrane-aligned:
- Rollout step $\leftrightarrow$ membrane evolving
- Naturally align with SNN



How can we leverage SNNs to solve PDEs effectively and efficiently?

S²MoE: SPIKING-BASED MOE FOR SCIENTIFIC MACHINE LEARNING

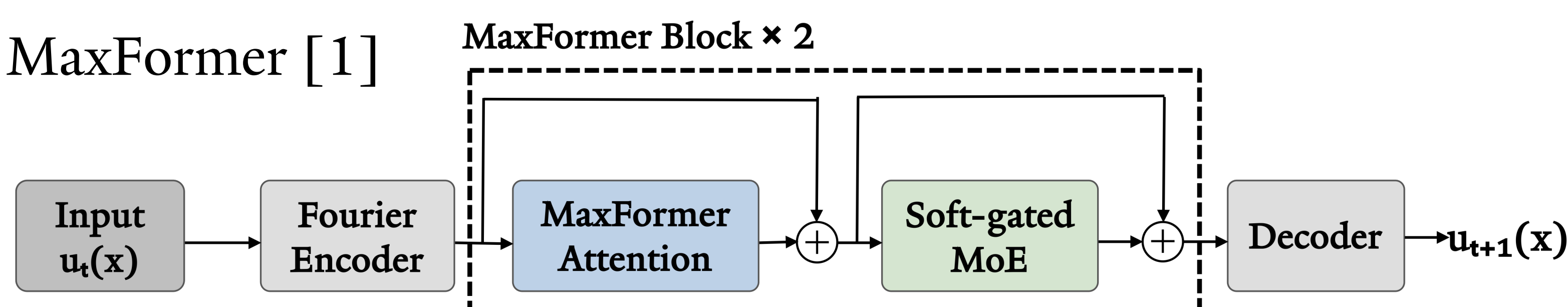
Shock & Smooth Parts



- Partial Differential Equation (PDE) solutions exhibit two regimes: **Shock & Smooth parts**, based on the gradient
- Specialize experts for shock & smooth regions

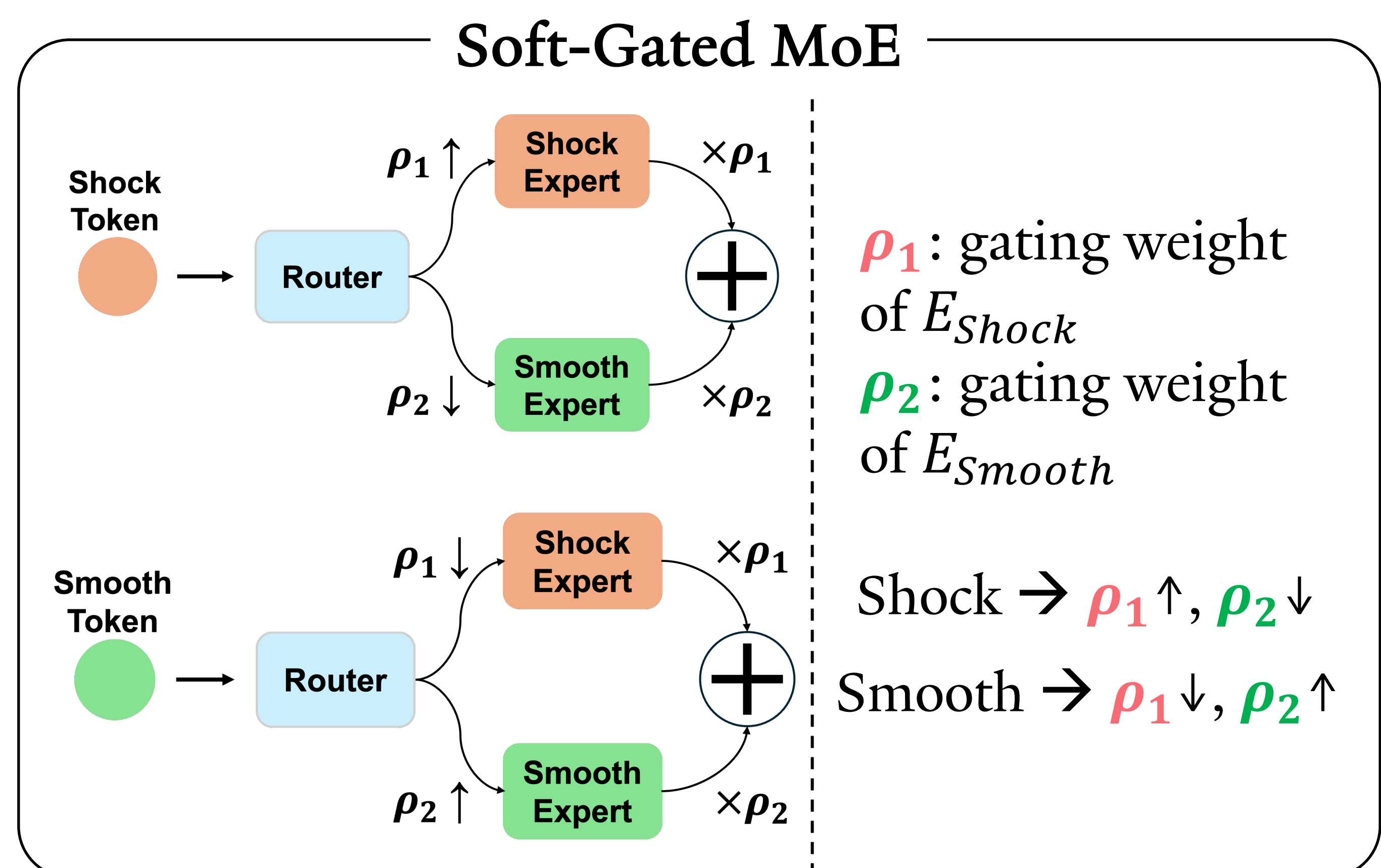
Baseline Architecture

- MaxFormer [1]



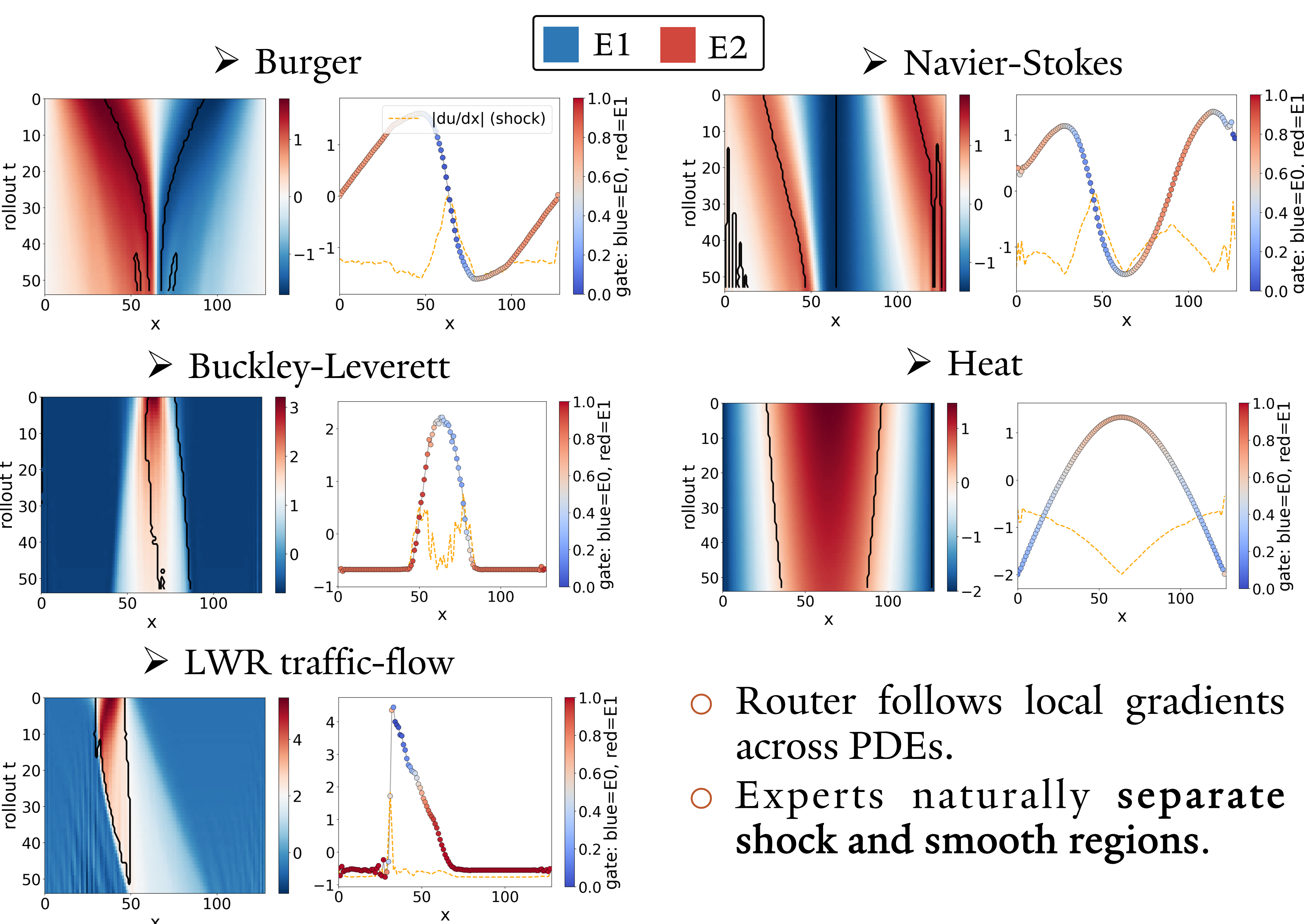
Soft-Gated Mixture-of-Experts

- Each expert specializes in shock or smooth regions
- All experts are activated for every token (soft gating, no top-k)



EXPERIMENTAL RESULTS

Physics-aware Routing

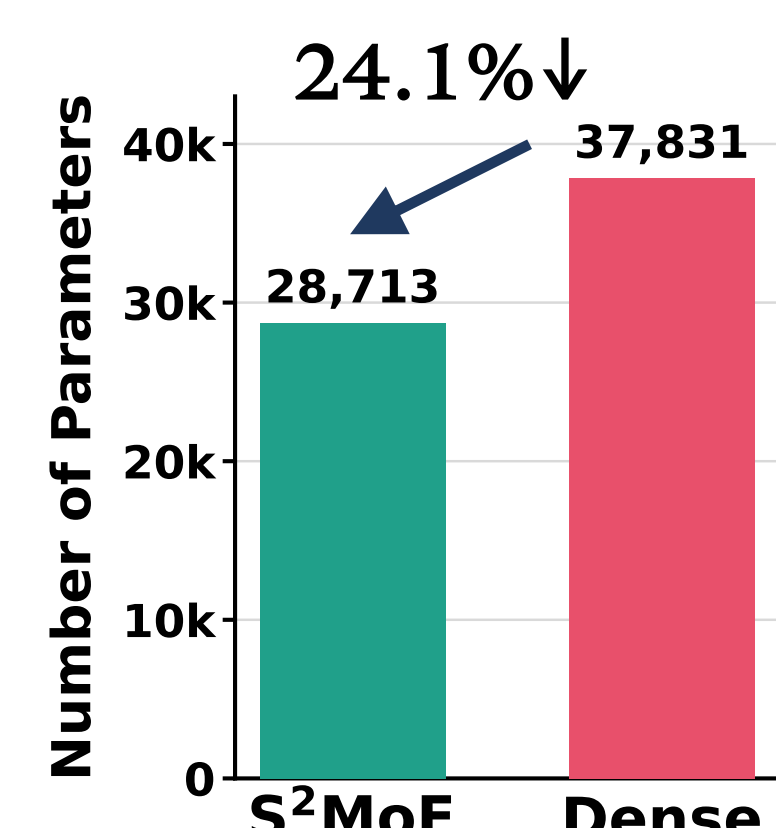


- Router follows local gradients across PDEs.
- Experts naturally separate shock and smooth regions.

Evaluation Results

- Relative L_2 error ($\downarrow$) on 5 benchmarks with lower parameters

Benchmark	MoE	Dense MLP
Burgers	0.111	0.144
Buckley-Leverett	0.075	0.094
Heat	0.133	0.133
Navier-Stokes	0.195	0.210
LWR	0.195	0.197



Hardware Efficiency Comparison

- Evaluated on Intel® Core™ Ultra 7 258V

